SpectraLab 1.3.1: measurement, analysis and provenance

Release documentation | 2026-10-27 | Christer Törnkvist

Scope and reading order

SpectraLab acquires spectra through an external instrument adapter, preserves immutable source archives, computes explicitly identified analyses and exports traceable reports and spectral artifacts. Standard/base MATLAB is the runtime policy. External Python and ArgyllCMS tools remain separate dependencies with their own licences.

This document describes release 1.3.1. The current sections take precedence over the retained historical technical foundation in the appendix. The appendix describes the older v1.0.1 architecture and is marked HISTORICAL on every page. Earlier version labels and numerical-method descriptions there must not be read as the current release contract.

Release 1.3.1 corrects density weighting, CRI rounding, reflectance units and Camera-41 export coverage. It includes portable Python preparation from the intervening development work. These corrections do not modify original measured archives, accepted Camera-41 projects or previously issued reports. Recalculation produces a separate reviewed revision.

Acquisition and immutable records

The verified acquisition family uses X-Rite i1Pro2 and GretagMacbeth Eye-One Pro through ArgyllCMS spotread. Preserve instrument identity, mode, calibration evidence, operator comments, wavelength grid, values, units, timestamps and archive provenance. Incomplete or calibration-only output must not become a completed measurement.

A durable archive preserves the actual measured data and its scientific content identity. A derived spectrum or analysis is a new record. A file checksum verifies bytes; a scientific content hash binds canonical scientific content. Neither alone establishes correct physical measurement geometry or instrument accuracy. New typed emission JSON also records a numeric datetime reconstruction residual, preserving low-order timestamp precision without executable object bytes; older packets remain readable.

Runtime and preparation

Each installation uses its own isolated Python 3.13.6 environment with pinned package versions. Prepare can obtain a private interpreter on macOS using a checksum-verified preparation tool; system Python is unchanged. Identify/check remain read-only. ARM is preferred on physical Apple Silicon. A complete installation and first-time internet access are required. See PYTHON_ENVIRONMENT.md.

Density calculations in 1.3.1

Separate transmission from source weighting

Let S(lambda) be the measured sample signal and R(lambda) the corresponding measured source/reference signal. ISO visual and Status A/M wrappers first form T(lambda) = S(lambda) / R(lambda) on a shared wavelength grid. Their documented spectral product Pi(lambda) already includes the required spectral weighting components.

The revised basis is D = -log10(integral(T * Pi) / integral(Pi)). This prevents the measured source spectrum from being applied twice. Revised outputs identify CalculationVersion=DENS-002 and IntegrationBasis=transmittance.

The generic weightedDensity function retains source-weighted calculation as its compatibility default: D = -log10(integral(S * W) / integral(R * W)). The wrappers accept an explicit source-weighted basis for diagnostic comparison. whiteDensity intentionally retains source-weighted V(lambda) weighting and is not ISO visual density. The two quantities must not be conflated.

Weight tables, integration and missing support

Documentary Status A/M log products are interpolated in log space on a 1 nm weighting grid before conversion to linear weights. Integration is trapezoidal; this is not ISO 5-3 Annex B weighting or ASTM E308 colourimetry. The implementation records incomplete weighting support and does not extrapolate missing measured wavelengths. Spectral calculation alone does not establish certified densitometer geometry or ISO conformity.

Dense samples and negative values

No automatic clipping threshold is invented. Negative signals outside the active weighted interval are excluded and their wavelengths recorded. Negative signals inside active weights require review and remain an error for the analysis. A zero reference under active weights is rejected. Rejection leaves the original measured archive intact.

A comparison of historical C-41 spectra is documented in SCIENTIFIC_REVIEW_20261026.md. The numerical changes are calculation revisions, not evidence that earlier physical measurements were changed or that downstream film profiles should be replaced automatically.

Colourimetry, spectral units and exchange

XYZ, Lab and reflectance units

Colourimetry combines measured reflectance and the chosen illuminant on common supported wavelengths, normalises the reference white to Y=100 and applies CIE 1976 Lab. Integration remains trapezoidal on supported spectral data; ASTM E308 weights are not implemented.

Reflectance units are explicit: percent values are converted to fractions once; fractional inputs are used directly. Ambiguous labels are rejected. A derived ColorChecker artifact labelled Unit=1 stores fractional values. Original-value archives and JSON exports remain their own records. XYZ and Lab are derived results, not replacements for measured patch spectra.

CRI rounding and emission metadata

Reported Ra is the rounded mean of the unrounded R1-R8 values. Individual reported indices may be rounded separately for display but are not averaged after rounding.

Future explicit spotread -e/-p spectral modes record mW/(m^2 sr nm); -a records mW/(m^2 nm). Unknown or relative modes retain an unknown scale. This is mode metadata and does not create a radiometric calibration or relabel historical measurements.

Camera-41 hand-off

Single/series transmission artifacts, ColorChecker artifacts and the qualitative wavelength colour bar use the same inclusive 400-730 nm contract. Partial coverage is rejected and no extrapolation is allowed. This exchange interval does not crop immutable source archives or globally limit CIE colourimetry to that interval.

Standalone JSON retains measured spectral content and available archive/provenance identities. Camera-41 owns camera calibration, RAW sampling, physical camera identity and the subsequent film-characterisation workflow. Each application verifies incoming content without rewriting the other's source measurements.

Verification, limitations and retained foundation

Release evidence

The scientific regression suite covers source-shape independence of transmission-weighted density, explicit compatibility weighting, zero/negative references, fraction/percent equivalence, CRI rounding, fixed export coverage, archive round trips, ColorChecker fractional artifacts and spotread mode units. The full release test result and package verification are recorded in releases/RELEASE_NOTES_v1.3.1.md and docs/release-validation-v1.3.1.json.

Automated tests use MATLAB R2025b and synthetic fixtures. No new physical instrument, printer, densitometer geometry or operating-system combination is qualified merely by these tests. Obtain authorized reference weights and validate measurement geometry and reference materials before claiming full standards conformance.

Historical appendix

The following retained technical foundation was issued for v1.0.1 on 2026-08-31. Its architecture discussion is useful background. Its release labels, formulas and numerical claims describe that historical implementation. The current 1.3.1 sections above define the corrected calculation and export behaviour.

The archived foundation is retained separately as docs/scientific-reports/foundation-v1.0.1-historical.pdf. Its original bytes are preserved. The combined current report adds a historical margin label to displayed appendix pages only.

Further reading

README.md provides installation and release links. USER_GUIDE.md and GETTING_STARTED.md describe the operator workflow. SCIENTIFIC_REVIEW_20261026.md records findings, a historical comparison and qualification limits. ED-017 describes the density framework and SPECTRAL_ARTIFACT.md describes artifact structure. Their historical decisions are read together with the 1.3.1 scientific review, not as new physical certification.

SpectraLab: Spectral Measurement, Analysis and Provenance

A base-MATLAB package for acquiring, preserving, analysing and communicating calibrated spectral measurements

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Purpose. This report describes SpectraLab as a scientific measurement package rather than as a plotting utility. It explains the chain from a physical spectrum and X-Rite spectrophotometer through ArgyllCMS acquisition, immutable archives, registered analysis, reports and controlled hand-off to other projects such as Camera-41.

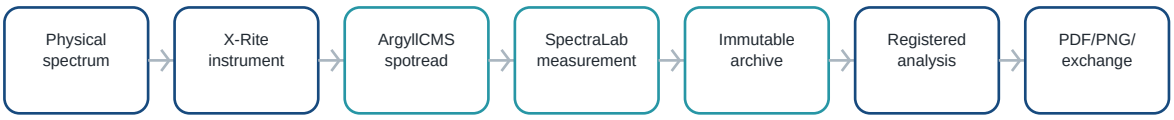


Figure 1. SpectraLab separates the physical instrument, acquisition backend, scientific record, derived analysis and presentation.

Document data	Value
Document status	Technical description
Stable release contract	SpectraLab v1.0.1
Reviewed source line	v1.0.2-dev patch-development worktree; not a published release
Author	Professor Christer Törnkvist
Report date	2026-08-31
Verified instrument workflows	X-Rite i1Pro2 and original GretagMacbeth Eye-One Pro Rev. B
Runtime policy	Standard/base MATLAB only; external acquisition adapters separately versioned

Executive summary

SpectraLab is an open-source MATLAB package for reliable spectral measurement. Its governing principle is "Measure once. Save forever. Verify always." A successful acquisition becomes an immutable scientific archive containing wavelength, measured values, units, instrument and calibration information, quality evidence and a deterministic content hash. Analyses are derived later and do not overwrite the measurement.

The verified acquisition path uses an X-Rite i1Pro2 or an original Eye-One Pro Rev. B through ArgyllCMS `spotread`. Python, `pexpect` and `ptyprocess` control the bounded external process. SpectraLab validates returned wavelength/value data before it creates an archive. Calibration output, incomplete output and failed measurements never become scientific measurement records.

The analysis layer supplies transmission, optical density, white and ISO visual density, Status A and Status M RGB density, spectral mean and difference, CIE XYZ, xyY, CIELAB and CIE 13.3 CRI. A canonical registry owns each reportable analysis, its input roles, result fields and figure contract. PDF, PNG and tabular outputs communicate derived results while preserving the original archive as authority.

SpectraLab does not turn a compatible instrument into a reference laboratory, guarantee traceability to a national metrology institute, or remove the need for correct calibration, geometry, warm-up, signal level and operator judgement. Its contribution is controlled acquisition, explicit processing, reproducible data identity and durable evidence.

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1. Purpose, scope and scientific boundaries

SpectraLab manages measured spectra as long-lived scientific records. It supports acquisition, archive validation and restoration, collection handling, reference/sample analysis, colour calculations, standard spectral weighting, plotting, PDF/PNG reporting and structured exchange. The software is designed so that measurement and scientific interpretation remain usable even when a particular user script or presentation changes.

The package is not itself a spectrophotometer. It does not directly sense radiation and does not supersede the calibration or uncertainty specification of the connected instrument. It accepts calibrated spectral results from the configured backend and records the context required to judge and reproduce subsequent calculations.

Layer	Owns	Does not own
Instrument	Physical detector, optics, internal calibration and raw response	SpectraLab archive policy
ArgyllCMS	Instrument communication, measurement mode and returned spectral result	SpectraLab scientific identity
SpectraLab	Workflow, validation, archive, analysis, report and provenance	National metrology traceability or sample preparation
Operator	Geometry, placement, warm-up, sample identity and scientific judgement	Silent modification of immutable measurements

SpectraLab distinguishes emission, reflectance and reference/sample calculations. Measurement meaning and units must be declared. A normalized spectral curve may describe shape, but it must not be presented as an absolute radiometric result unless the acquisition supplies absolute units and the complete geometry supports that interpretation.

2. System architecture and responsibility separation

User workflows	Work/examples and public API
Reporting	Registry, document model, vector PDF and 1400 x 700 PNG
Analysis	Transmission, density, Status A/M/W, XYZ, Lab, CRI, mean and difference
Archive and I/O	SLAB-MAT, SLAB-JSON, content hash, CSV/TXT and collections
Core model	Spectrum, filter, session, calibration, instrument
Driver boundary	ArgyllCMS spotread process control and parsing

Figure 2. Layered architecture. Higher layers consume explicit results from lower layers rather than reaching through them.

The public API expresses measurement or analysis intent. Drivers hide backend commands. The core model represents spectra and filters. Archives convert object state into stable structures suitable for long-term preservation. Analysis functions consume spectra or restored archives and return explicit result structures. Report generation accepts only registered analyses and verified input roles.

This separation prevents an export routine from creating new scientific meaning, an example script from inventing a competing formula, or a plotting function from silently calculating colour values. Derived information is attached as a new product and never replaces the measurement that supports it.

Architectural rule	Consequence
Measure once	Original spectral values are preserved and re-analysed later
Archive independence	Stable structures reduce dependence on MATLAB class implementation
Registered analysis	One authoritative definition of inputs, outputs, method and figure
Presentation separation	PNG information panels and PDF results cannot redefine calculations
Explicit hand-off	Downstream projects receive hash-bound read-only evidence

3. Instrument acquisition and calibration authority

The current verified workflow uses ArgyllCMS `spotread` with the X-Rite i1Pro2. Post-release examples have also been physically verified with an original GretagMacbeth Eye-One Pro Rev. B, identified as `i1Pro`, in standard and optional high-resolution modes. Generic `spotread` selection currently resolves to the i1Pro2 compatibility path; the explicit instrument identity is preferred.

In automatic mode, calibration and measurement are bounded one-shot processes using the documented Spotread `-O` operation. SpectraLab waits for a recognised state, asks the operator to place the instrument correctly, runs one operation, parses the output and accepts a spectrum only after validation. A repeated calibration request receives one controlled recovery cycle rather than an unlimited retry loop.

State or event	SpectraLab response
Calibration required	Request white-reference placement; run and verify calibration
Calibration succeeds	Record provenance; request sample/source placement
Complete spectral measurement	Validate samples, metadata and command outcome; create archive
Calibration-only or incomplete output	Discard as measurement; create no archive
USB, timeout or backend failure	Abort bounded operation and series; report structured error

ArgyllCMS and the physical instrument are authoritative for calibration status. SpectraLab may record elapsed time, but it cannot override a calibration request reported by the backend. Correct placement, matching white reference, clean optics and sufficient warm-up remain operator responsibilities.

Development boundary. The architecture can accept further drivers, but no other instrument shall be described as verified merely because ArgyllCMS lists it. Each new workflow requires explicit integration tests and physical verification.

4. Spectrum and archive data model

A spectrum consists of a strictly increasing wavelength vector in nanometres and one corresponding finite value per wavelength. Its label, units, timestamp and metadata describe scientific meaning. Collections group spectra without merging their identity. A measurement result also carries instrument, calibration, command and quality information.

The canonical MAT archive stores stable structures rather than requiring the original class implementation. Its UUID identifies one archive instance. Its SHA-256 content hash identifies deterministic scientific content, including measured values, wavelengths, instrument and quality information. Volatile archive time, filename and editable comments do not alter scientific content identity.

Identity or record	Meaning
Archive UUID	One particular archive instance
ContentHash	Deterministic scientific payload identity
Metadata	Operator, project, sample, comments and declared context
Quality	Acquisition and validation evidence retained with measurement
Processing history	Derived-operation provenance; never a replacement for source values

SpectraLab also defines a readable `.slab.json` format with schema `spectralab.spectrum.v1`. CSV and text are communication formats; they do not carry the complete authority of the native archive unless the receiving contract explicitly preserves equivalent metadata and identity.

5. Wavelength alignment, filtering and integration

Reference/sample calculations require a common wavelength domain. Identical grids are required by default. If grids differ, the user must explicitly enable resampling. SpectraLab restricts processing to the common measured interval and performs no silent extrapolation. The default refinement factor is four and the default interpolation method is shape-preserving cubic interpolation (`pchip`); `makima` and spline are explicit alternatives. Every original wavelength remains in the refined grid.

A `SpectralFilter` represents a wavelength-dependent weighting function. It may be tabulated or analytical, but it has an explicit valid range and returns one weight per requested wavelength. Weighted responses remain inspectable as functions of wavelength; scalar integration is a later operation, normally performed with trapezoidal numerical integration.

Operation	Scientific control
Transmission	Sample/reference ratio on an explicitly aligned grid
Optical density	-log10 of positive transmission with defined validity checks
Weighted density	Reference and sample integrated under a named spectral product
XYZ	Reflectance or SPD integrated with declared illuminant and CIE colour-matching functions
Mean/difference	Aligned source set with method, range and source identities recorded

Interpolated points do not create new measured information. They support numerical integration and comparison within the observed domain. Reports therefore record whether resampling occurred, the method, refinement factor and effective wavelength range.

6. Registered scientific analyses

The canonical analysis registry is the sole authority for reportable analyses. Each entry declares an analysis ID, public name, method, standard, definition version, ordered input roles, result fields and whether a figure exists. Unknown IDs and incomplete or contradictory definitions fail before report creation.

ID	Analysis	Inputs	Figure
ANL-SPECTRUM	Measured spectrum	Measurement	Yes
ANL-CRI	CIE colour rendering index	Measurement	Yes
ANL-001	Transmission	Reference, sample	Yes
ANL-002	Optical density	Reference, sample	Yes
ANL-004	White density	Reference, sample	No
ANL-005	Status A RGB plus visual W	Reference, sample	No
ANL-007	ISO visual density	Reference, sample	No
ANL-008	Status M RGB plus visual W	Reference, sample	No
ANL-009	Spectral arithmetic mean	Source set	Yes
ANL-010	Signed spectral difference	A, B	Yes

Separate public functions provide CIE XYZ, xyY and CIELAB calculations and ColorChecker aggregate colorimetry. The registry verifies that every declared scalar result exists after execution. Mean may create a traceable derived MAT archive; difference intentionally creates no derived measurement archive.

Standards note. Status A/M weighting tables are documented from ANSI/ISO 5-3:1995 Table 3, while ISO 5-3:2009 is the current standard. This report does not claim an unqualified certification of equivalence between editions; the exact bundled data source and hash remain authoritative for reproducibility.

7. Reporting, plots and presentation rules

SpectraLab uses one graphical presentation architecture for interactive figures, PNG exports and PDF report figures. The canonical interactive/PNG geometry is 1400 x 700 pixels with spectral axes on the left and legend/information space on the right. Wavelength is in nanometres and a spectral colour guide may appear along the lower axis. Percent-valued transmission and reflectance use 0-100 percent by default; emission retains its recorded unit and scale.

Interactive figures and PNG files may show concise measurement information such as peak, integral, CCT, Duv, CRI, project, operator and instrument. A PDF figure contains the plot, title and legend, while authoritative numerical values and provenance appear in the report text and tables. This prevents a report from duplicating or clipping a screen-oriented information box.

Output	Purpose	Authority
Interactive figure	Immediate operator inspection	Visual aid
PNG	Full-resolution portable figure with measurement context	Presentation derivative
PDF	Self-contained analysis, results, quality and provenance	Human-readable report
MAT/JSON archive	Measured values and complete machine-readable identity	Scientific source record

Report headers, footers, filenames, page framing and minimum margins are part of the controlled presentation. A visually correct plot is still not scientific proof: values must originate from registered result fields and source hashes must remain valid.

8. ColorChecker measurement and correction workflow

A ColorChecker session is a JSON manifest plus one immutable reflectance archive per patch. A controlled target definition can lock the manufacturer, model, geometry, patch count, orientation and nominal target identity. The first controlled definition is the 10 by 14 X-Rite ColorChecker Digital SG with 140 patches and row-major acquisition from A1.

The session records calibration events, instrument identity, resolution, patch coordinate, archive filename, UUID and content hash. Acquisition creates neither report nor plot automatically. Session reports, patch reports, aggregate colorimetry and CSV/MAT exports are explicit later actions. An erroneous patch is not overwritten: controlled remeasurement creates replacement archives and an amendment manifest while preserving every original record.

Stage	Persistent result
Target assignment	Embedded controlled definition and SHA-256

Stage	Persistent result
Patch measurement	One immutable SLAB-MAT reflectance archive per patch
Session state	Revisioned JSON manifest with archive identities
Remeasurement	Separate amendment and replacement archives
Colorimetry	Derived D50/declared-illuminant XYZ and Lab JSON/CSV/PDF

The physical measured reflectance R(lambda) is primary. Derived XYZ or Lab is not a substitute for the archived spectra. This allows Camera-41 or another consumer to recalculate colour under a different declared illuminant or observer without remeasuring the target.

9. Provenance, reproducibility and integrity

SpectraLab treats provenance as part of the scientific result. Measurement archives retain instrument name and serial, calibration state, operator, timestamp, measurement kind, mode, resolution, command/backend information, wavelengths, values, units and quality. Reports add source filenames and content hashes. Derived products record analysis identity, definition version, alignment and interpolation parameters, wavelength range and source records.

Threat	Control
File renamed or copied	Scientific ContentHash remains tied to payload, not filename
Archive modified	Validation recomputes and compares canonical SHA-256
Patch corrected	Original retained; amendment creates new identities
Analysis changed	Analysis ID, definition version and source hashes identify calculation
Different wavelength grids	Default failure or explicit recorded resampling
Presentation changed	Scientific archive and registered results remain separate

Reproducibility means that the same archived scientific content and declared processing should produce the same numerical result within the stated computational tolerance. It does not mean that a new physical measurement must be numerically identical: instrument repeatability, positioning, sample variation, source stability and environment remain experimental variables.

10. Base MATLAB and external dependencies

Standard/base MATLAB is the only permitted MathWorks runtime product. Report Generator and other add-on toolboxes are forbidden dependencies. SpectraLab creates plots and vector PDF reports through ordinary base-MATLAB figures and `exportgraphics`. A regression test checks the source tree and MATLAB product dependency analysis.

External components are permitted because they are not MathWorks products and have explicit, separately versioned responsibilities. ArgyllCMS communicates with the instrument. Python plus `pexpect` and `ptyprocess` provide bounded process interaction. SpectraLab setup checks their presence and reports the resolved executables and versions.

Component	Verified/recommended release environment	Responsibility
MATLAB	R2024b or later; base product only	Core, analysis, archive, plot and report
Python	3.10 or later	External-process adapter runtime
pexpect	4.9 or later	Bounded Spotread interaction
ptyprocess	0.7 or later	Pseudo-terminal dependency
ArgyllCMS	3.5 or later	Instrument backend and spectral acquisition

A dependency is not hidden merely because it is external. The resolved path, version, command options and returned result belong in the acquisition provenance. A future backend must preserve the same scientific and failure-state contract.

11. Quality, limitations and development items

Regression tests cover spectrum and filter validation, archive round trips and content hashes, analysis formulas, resampling, standard datasets, registry invariants, PDF/PNG structure, ColorChecker sessions, correction provenance, public examples and the base-MATLAB rule. Practical use has additionally exercised i1Pro2 and Eye-One Pro workflows. Passing tests demonstrates consistency with the implemented contract; it does not prove absence of defects or quantify every physical uncertainty.

Limitation	Required interpretation
Instrument accuracy	Use manufacturer specifications, calibration and independent checks
Geometry and stray light	Control physically; software cannot reconstruct missing measurement validity
Absolute radiometry	Do not infer from arbitrary or normalized units

Limitation	Required interpretation
Interpolation	Numerical aid within measured domain, not additional observation
CRI	CIE Ra remains supported but does not describe every aspect of colour quality
Downstream decisions	Remain the responsibility of the user or receiving project

Items requiring future development or explicit decision

- Physically verify and document each additional instrument/backend before claiming support.

- Decide whether later releases add a modern colour-fidelity metric alongside CIE Ra; do not silently reinterpret existing ANL-CRI records.

- Maintain a traceable standards review for Status A/M source tables and any difference between legacy documentary data and current ISO editions.

- Extend uncertainty reporting only when its physical and statistical assumptions are defined; do not label repeatability alone as total measurement uncertainty.

- Keep v1.0.2-dev corrections compatible with the accepted v1.0.1 scientific and archive contracts unless a formally versioned migration is approved.

Conclusion. SpectraLab contributes an auditable chain from a calibrated instrument result to a durable spectrum, explicit analysis and readable report. Its scientific value lies in preserving measured information, declaring every transformation and refusing incomplete evidence - not in replacing the instrument or the operator.

Appendix A. Public function and record map

Function or group	Role
spectralab.drivers.createInstrument	Creates an explicit verified instrument adapter
spectralab.core.Session	Controls open, calibration and measurement lifecycle
spectralab.measurement.oneShot	One bounded calibration/measurement operation
spectralab.archive.create/save/load/restore/validate	Creates and verifies immutable scientific archives
spectralab.core.resampleSpectrum/Pair	Explicit in-domain wavelength refinement and alignment
spectralab.analysis.*	Named scientific calculations with explicit results
spectralab.report.listAnalyses/describeAnalysis	Public inspection of canonical registry
spectralab.report.generate	Registered analysis to PDF/PNG/results
spectralab.colorchecker.*	Controlled target sessions, corrections and colorimetry
spectralab.io.*	SLAB-JSON, CSV, TXT, collection and colorimetry exchange

Error handling

Public failures use structured SpectraLab identifiers and abort before writing a trusted output. A failed measurement is not represented by an empty archive. Existing immutable archives are not overwritten by correction workflows.

Appendix B. Camera-41 hand-off

SpectraLab and Camera-41 have separate hash-bound responsibilities. SpectraLab characterises the physical target and illumination. Camera-41 characterises the photographed RAW target, camera response, spatial capture quality and negative inversion. Neither rewrites the other application's source records.

SpectraLab supplies read-only	Camera-41 adds
Target-defined ColorChecker session JSON	Camera-specific imaging-role map
Target definition and SHA-256	RAW target orientation and patch geometry
One reflectance archive per patch	Linear camera-RGB patch measurements and masks
Patch UUID and ContentHash	Clipping, S/N, spatial uniformity and profile fit
Illuminant spectrum archive	RAW-to-XYZ/ACEScg model and residuals
Correction and remeasurement provenance	Output profile, inversion and TIFF/OpenEXR provenance

Camera-41 recalculates reference XYZ from measured R(lambda) and the selected illuminant when required. Existing derived Lab/XYZ is useful for inspection but never replaces the spectral source. Image-specific roles are versioned by Camera-41 and are not retrospectively written into SpectraLab patch archives.

Appendix C. References

[1] SpectraLab project repository and releases. https://github.com/chto0703009/SpectraLab. Stable release v1.0.1.

[2] SpectraLab source documentation: README, architecture, file format, analysis registry, Spotread one-shot workflow, graphical profile, ColorChecker session architecture and Camera-41 hand-off contract.

[3] X-Rite. *i1Pro 2 spectrophotometer*, archived product literature. X-Rite i1Pro 2 product literature.

[4] ArgyllCMS. *spotread* documentation and operation of supported instruments. https://www.argyllcms.com/doc/spotread.html; https://www.argyllcms.com/doc/instruments.html.

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[7] CIE. *CIE 013.3-1995 Method of measuring and specifying colour rendering properties of light sources*. CIE publication page.

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[9] The MathWorks, Inc. (2025). *MATLAB* [computer software]. Natick, Massachusetts, United States. https://www.mathworks.com/products/matlab.html.

[10] Python Software Foundation. *Python* [computer software]. https://www.python.org/.

[11] Camera-41 project repository and hand-off consumer. https://github.com/chto0703009/Camera-41. Access depends on repository permissions.

Document conclusion. SpectraLab is a controlled scientific-data system built around explicit measurement identity, immutable archives, registered analysis and reproducible reporting. The verified X-Rite/ArgyllCMS path provides spectral data; SpectraLab preserves and interprets those data without concealing the physical and metrological limits of the measurement.